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Big-Data for Computational Finance

Assignment 1: Report

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Introduction

The Markowitz model is a fundamental concept in modern portfolio theory [1]. This model, which is also known as the Mean-Variance Optimization model, helps investors build portfolios that aim to either maximize the expected return for a specific level of risk or minimize the risk for a certain level of expected return. In other words, each of the situations could be handled. By analyzing the trade-off between risk and return, the Markowitz model provides a systematic way to achieve the best possible investment outcomes based on individual risk preferences, that is mean the investors will make the preferable choice, depending on their desire to take the risk [2].

Definition of the problem

The Markowitz model is designed to minimize the risk (variance) of a portfolio while meeting certain target-return taking in consideration constraints. It does this by solving a mathematical problem called quadratic optimization. There is different type of Constraints, for instance expected Return, in which the portfolio should achieve a specific target return, total investment, in which the sum of the weights of all assets should equal 100% of the investment and no short selling, in which the weights of the assets should be non-negative (we can't invest a negative amount in any asset). In simpler terms, the Markowitz model helps us find the best mix of assets that keeps our risk low while still aiming for a desired return.

$$\begin{aligned} &\text{minimize } \mathbf{x}^T \mathbf{C} \mathbf{x} \\ &\text{subject to } \boldsymbol{\mu}^T \mathbf{x} = r, \\ &\quad \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Where:

- $\mathbf{x}$ is the vector of asset allocations.
- $\boldsymbol{\mu}$ is the vector of expected returns.
- $\mathbf{C}$ is the covariance matrix.
- r is the target return.

Task 1: Standard Markowitz Optimization

Aim

Markowitz optimization in task 1, the aim is to minimize the portfolio variance with all capital invested and the expected return equals r and we are given a set of expected return $r = \{\}$ the other constrain is the sum of asset weights equals 1 that correspond to the whole capital should be invested. In task 1, Markowitz optimization problem try to find the optimal weights $\mathbf{x}$ for a portfolio of assets. for instance, x_1 x_2 x_3 that will be invested in the asset which minimizes the risk while achieving a specified target return r . This engaging solving the following quadratic programming problem:

$$\begin{aligned} &\text{minimize } \mathbf{x}^T \mathbf{C} \mathbf{x} \\ &\text{subject to } \boldsymbol{\mu}^T \mathbf{x} = r, \\ &\quad \mathbf{e}^T \mathbf{x} = 1, \\ &\quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

Where:

r is the target return.

$\mathbf{e}$ is a vector of ones. This constrain mean the whole capital will be invested.

Finding

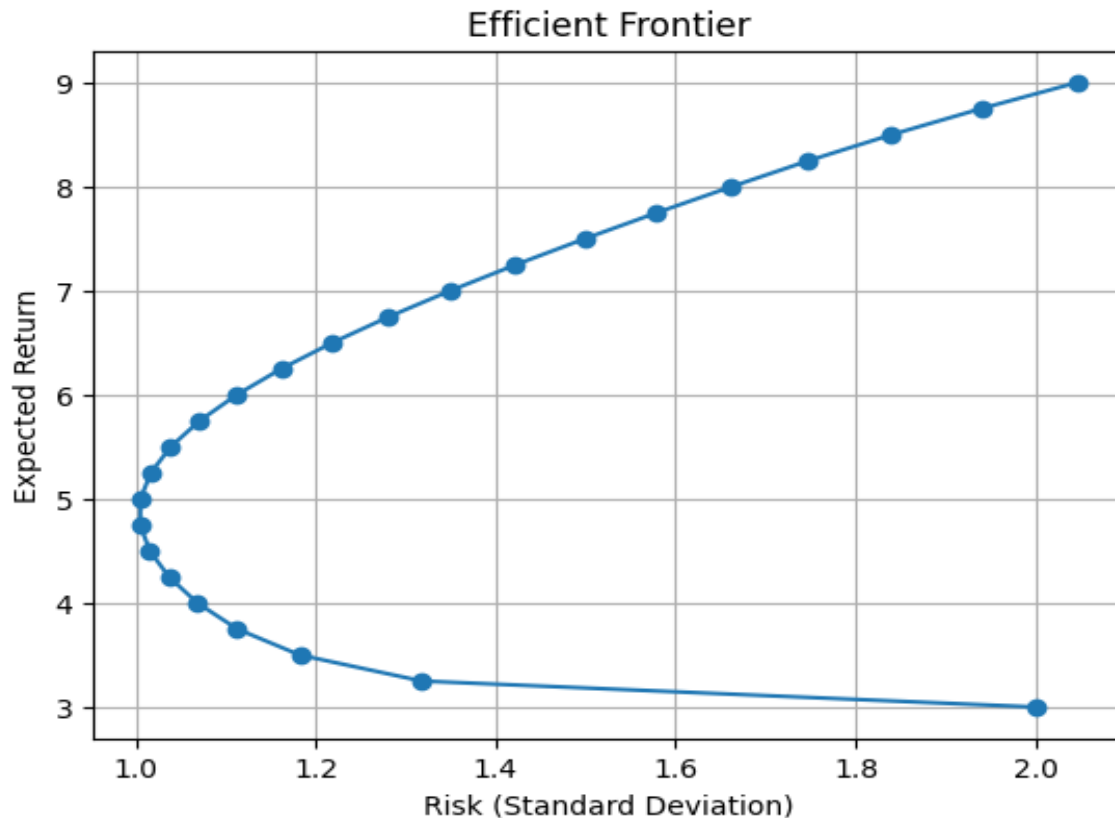


Figure 1: The plot illustrates the efficient frontier for 29 different values of r ranging from 2.00 to 9.00.

Explanation:

For each target return (r) from 2.00 to 9.00 in increments of 0.25, The plot shows these 29 pairs, where each point $(\sigma(x), \mu(x))$ represents the risk-return trade-off for a specific target return. This plot is known as the "Efficient Frontier" in portfolio theory. For each target return from 2 to 9 the code solves a quadratic programming (QP) problem. The goal is to find the portfolio weights $x_1, x_2, \dots$ that minimize risk (variance) for a given target return, subject to the constraints that the portfolio has, which is investing the whole capital. This plot shows the risk and return of a portfolio change as we vary the target return (r). By solving the quadratic programming problem iteratively and plotting the results, we gain insights into the optimal portfolio allocations under different risk-return scenarios. This analysis helps investors make informed decisions about portfolio construction based on their risk tolerance and return objectives. Therefore, as we can see that high return associated with high risk for example at return 9 the variance is high. were as at return 5 the risk is extremely low. So, if the inventor wants to get a moderate return with minimum risk, they can go for that option. and I notice from the graph that at return 2 the risk also is high that mean it the worst scenario. So, the main idea is to allocate the set of fractions $x_1, x_2, \dots$ in the asset to get a specific return and with the minimum risk. Solving the quadratic programming problem for different target returns, the code identifies the best possible combinations of assets (portfolios), helping investors make informed decisions about where to allocate their investments. When we go right the risk increases and when left the risk decreases.

Task 2: Allowing Partial Investment

Aim

To modify the standard Markowitz portfolio optimization problem to allow partial investment, where the sum of asset weights can be less than or equal to 1, we adjust the constraints while keeping the objective and other conditions the same, the target return required. In practical situations, investors don't always need to use all their funds for investments. This modification to the standard Markowitz model allows for the option to keep some capital in cash or low-risk assets. This is done by loosening the requirement that all available funds must be invested. Instead, the model now allows for the possibility of holding back some portion of the capital, providing flexibility in portfolio allocation decisions. This adjustment recognizes that investors may want to maintain liquidity or manage risk differently, apart from fully committing all available resources into the market. This engaging solving the following quadratic programming problem:

$$\begin{aligned} &\text{minimize } \mathbf{x}^T \mathbf{C} \mathbf{x} \\ &\text{subject to } \boldsymbol{\mu}^T \mathbf{x} = r, \\ &\quad \mathbf{e}^T \mathbf{x} \leq 1, \\ &\quad \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Findings

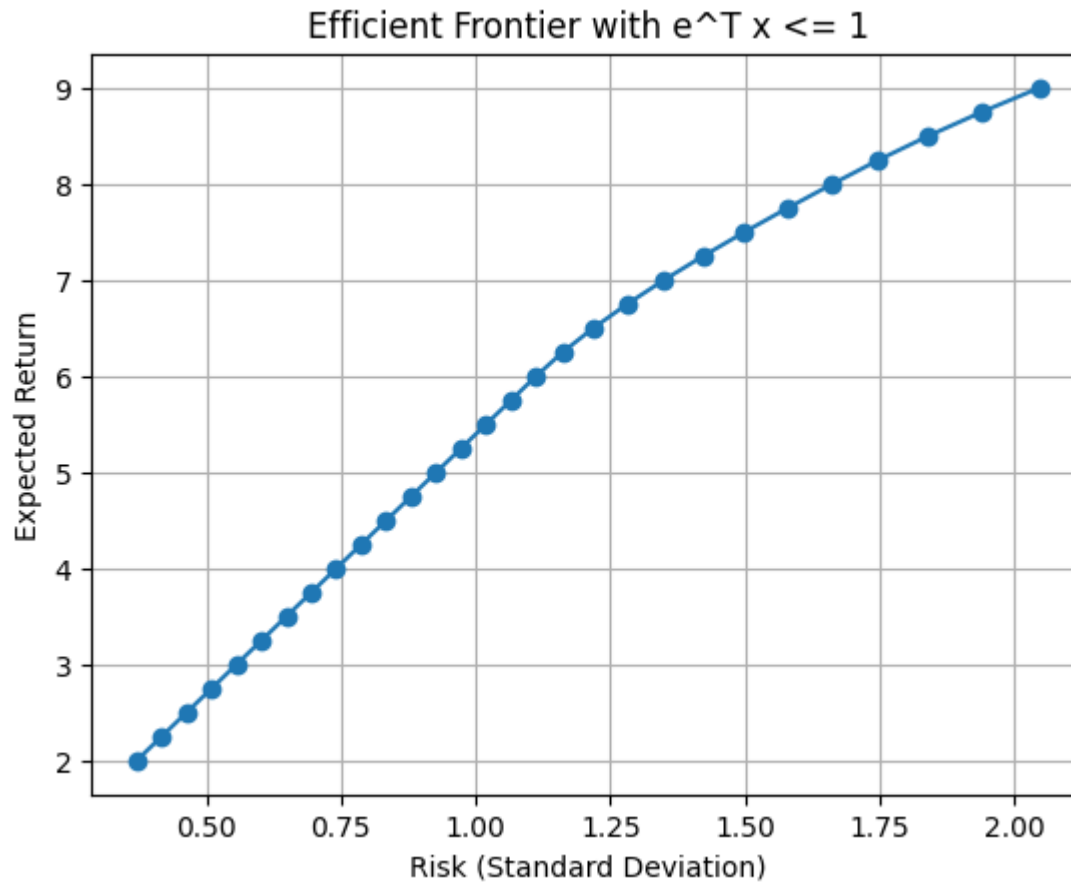


Figure 2: The plot is the efficient frontier under the constraint of not investing the whole capital.

Explanation

In Task 1, the goal was to construct an efficient frontier for a portfolio where the entire capital must be invested. This means that the sum of the weights of all assets in the portfolio (x) must equal 1. In Task 2, the modification allows for the possibility that not all capital needs to be invested. This means that the sum of the weight of the assets can be less than or equal to 1. This means that any leftover capital, which is not invested, does not earn any return and carries no risk. Allowing not all capital to be invested can be more realistic because investors might not always find it optimal to invest all their funds, especially if they can't find enough favorable investment opportunities. This figure is more realistic than the previous one. The efficient frontier shows lower risk points because not all capital has to be invested, potentially leaving some in a risk-free. I believe that the figure 2 is better because for example if we have an investor who is risk-averse or wants the option to keep some capital safe, this will provide more flexibility and can lead to lower overall portfolio risk because uninvested capital carries no risk. Therefore, we can notice here there is an almost linear relation between the expected return and the risk, the higher the return the higher the risk. we can notice from the graph that the low return with high variance is removed.

Task 3: loosen the Expected Return Constraint

Aim

In this modified version of the Markowitz model, we adjust the expected return constraint to ensure the portfolio achieves at least a specified return level r , or potentially higher. This change allows investors to aim for a minimum desired return without requiring an exact target. It offers flexibility by ensuring that the portfolio's expected return meets or exceeds r , accommodating varying market conditions and investor preferences for achieving certain levels of profitability. This modification enables investors to pursue their financial goals with greater adaptability in portfolio management, ensuring the portfolio performs satisfactorily relative to their return expectations. This task modifies the expected return constraint to allow for returns that are greater than or equal to r . This engaging solving the following quadratic programming problem:

$$\begin{aligned} &\text{minimize } \mathbf{x}^T C \mathbf{x} \\ &\text{subject to } \boldsymbol{\mu}^T \mathbf{x} \geq r, \\ &\quad \mathbf{e}^T \mathbf{x} = 1, \\ &\quad \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Finding

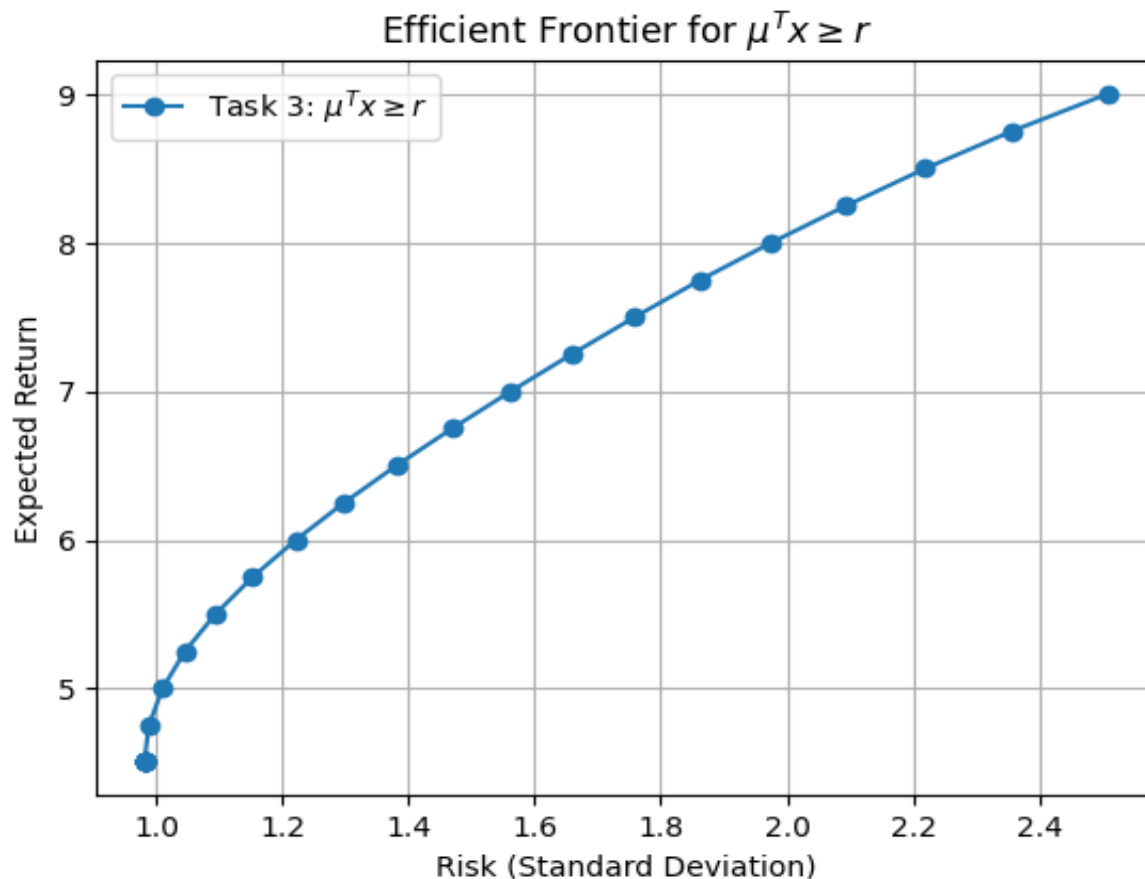


Figure 3: The plot below demonstrates the efficient frontier under the loosen return.

Explanation

In this task, the portfolio should achieve at least the target return, or it can exceed it. It introduces more flexibility, allowing the portfolio to potentially exceed the target return if it helps in minimizing risk. The portfolio must achieve the exact target return in task1 whereas it can achieve returns greater than or equal to the target return. With the modified constraint, the optimizer can choose portfolios that exceed the target return, potentially finding combinations that have lower risk for the same or higher return. Therefore, Task 1 Less flexible, as it must meet the exact target return. Task 3 More flexible, as it can exceed the target return. I believe this will provide better investment opportunities for the investor. Moreover, by looking at the graph, we notice that the high risk with the low return was removed and that is a better and make an advantage over task1. That means, for the same risk the investor goes for the higher return. And that is extremely important and benefic

Task 4: Allowing Short Selling

Aim

When we modify the Markowitz portfolio optimization to allow for short selling, we're essentially removing the rule that limits how we can invest in assets. Normally, this rule says we can only buy assets—we can't sell them unless we already own them. Short selling changes that. It lets us borrow assets we don't own, sell them right away, and then buy them back later (hopefully at a lower price). This flexibility can potentially increase the opportunities for profit because it allows us to benefit from both rising and falling prices in the market. This task demonstrates allowing short selling by removing the non-negativity constraint on the asset weights. This engaging solving the following quadratic programming problem:

$$\begin{aligned} &\text{minimize } \mathbf{x}^T \mathbf{C} \mathbf{x} \\ &\text{subject to } \boldsymbol{\mu}^T \mathbf{x} = r, \\ &\quad \mathbf{e}^T \mathbf{x} = 1. \end{aligned}$$

Finding

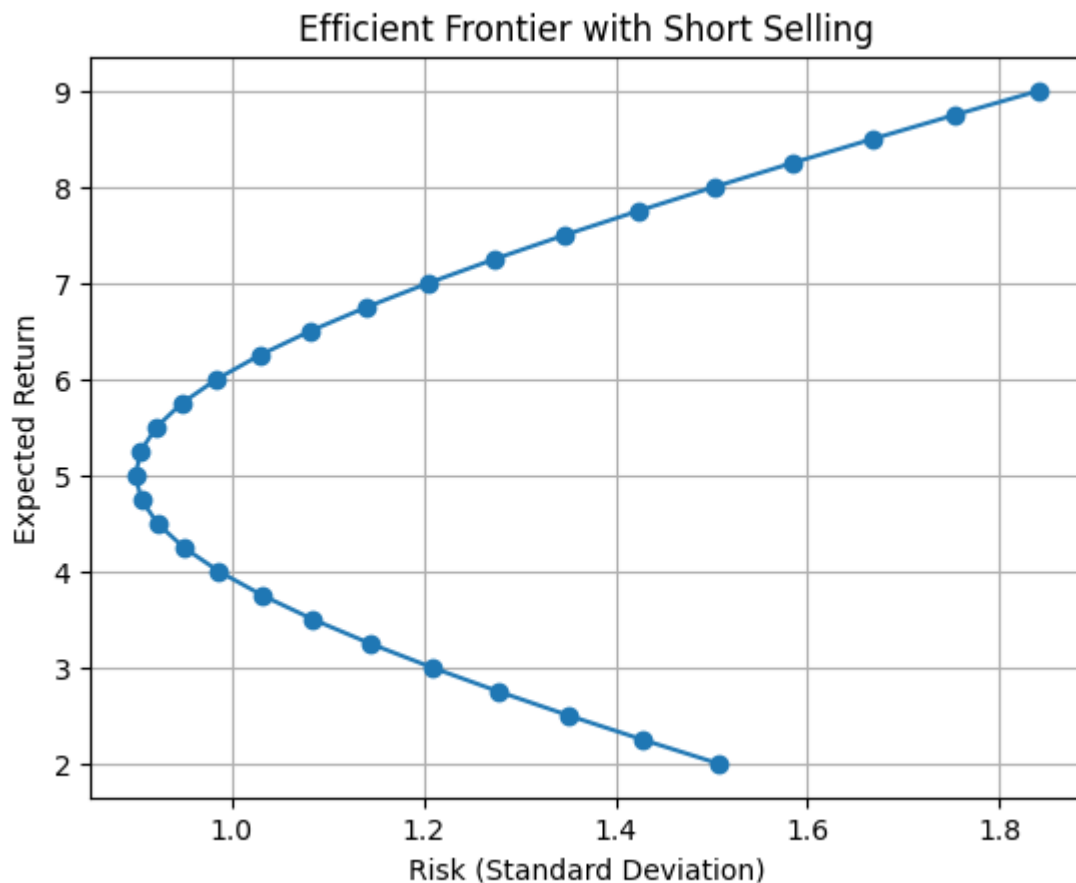


Figure 4: The plot below illustrates the efficient frontier under the short selling constrain.

Explanation

When we allow short selling in the Markowitz portfolio optimization, the efficient frontier—the set of optimal portfolios—becomes symmetric. This means it extends into both positive and negative return regions. This shows how short selling adds flexibility and risk to the portfolio strategy. Short selling allows for negative weights, meaning we can take on leveraged positions where we borrow assets to sell them, aiming to buy them back at a lower price. This capability expands the potential for both higher returns and higher risks. In contrast to scenarios without short selling (like in Task 1), where portfolios only aim for positive returns, allowing short selling means portfolios can now potentially achieve returns above or below zero. This demonstrates the broader range of risk and return possibilities available when short selling is permitted in portfolio management.

Conclusion

This report has examined several changes to the traditional Markowitz model used for portfolio optimization. Each change affects the efficient frontier, showing how risk and return trade-offs vary with different investment rules. The findings emphasize how adaptable the Markowitz model is, making it suitable for various investment situations and risk preferences.

References

- [1] Kamil, A.A., Fei, C.Y. and Kok, L.K. (2006). Portfolio analysis based on Markowitz model. *Journal of Statistics and Management Systems*, 9(3), pp.519–536. doi: <https://doi.org/10.1080/09720510.2006.10701221>.
- [2] Pandey, M. (2012). APPLICATION OF MARKOWITZ MODEL IN ANALYSING RISK AND RETURN A CASE STUDY OF BSE STOCK. *Risk Governance and Control: Financial Markets & Institutions*. [online] Available at: <https://virtusinterpress.org/APPLICATION-OF-MARKOWITZ-MODEL-IN.html> [Accessed 27 Jun. 2024].